A Novel Approach to Cipher Decryption Using Unified Multi-Cipher GAN Model

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Cryptography is the ultimate form of non-violent direct action.

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Abstract

Nowadays, the application of deep learning has significantly increased in cryptanalysis. The state-of-the-art model Unified Cipher Generative Adversarial Network model (UC-GAN), an unsupervised deep-learning neural network, has demonstrated its capability to translate a single plaintext into multiple ciphertexts and vice versa, using a single unified generator and a discriminator within the model. This model has been effective in parallel decryption of basic substitution ciphers. However, its scope is limited to simple, traditional substitution ciphers. In response, we first build a CatBoost-based supervised machine-learning distinguisher to evaluate if the distinguisher can do the multiclass classification successfully on different ciphers within the same cipher type, either substitution, transposition, or rotor cipher type. The distinguisher achieves an accuracy of around 90% for all cipher types. Therefore, the CatBoost-based distinguisher successfully classifies the ciphertexts into four classes, namely the fake ciphertext and the ciphertexts from the first, second, and third ciphers. Additionally, we propose the Unified Multi-Cipher Generative Adversarial Network model (UMC-GAN) in this paper. This model extends its capabilities beyond classical substitution ciphers to include transposition and rotor ciphers, which were prevalent in securing communications during World War II. We assess if the deep learning UMC-GAN model can generate realistic ciphertexts. It achieves an accuracy of nearly 98% by learning only data with no pre-existing knowledge. Our experimental results validate that the UMC-GAN, an unsupervised deep-learning GAN-based model, generates realistic ciphertexts within the same cipher type, and can fool the discriminator. These research findings suggest the potential of the UMC-GAN model in decrypting both block ciphers and practical encryption systems, such as those utilized by Telegram and WhatsApp.

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Chapter 1

Introduction

In an era where ensuring digital security is of utmost importance, the pursuit of resilient security algorithms has never been more crucial. Our research focuses on cipher decryption via a variant of the deep learning generative adversarial networks (GAN) model, named Unified Cipher Generative Adversarial Network (UC-GAN) [14].

Motivated by the need to improve upon shortcomings found in previous research, prompting us to conduct a thorough review of existing deep learning GAN-based cryptanalysis models. During this research, a significant limitation became evident: previous studies predominantly tested the existing popular GAN-based cryptanalysis models, the improved Pix2Pix model [2] with their embedding technique, the CipherGAN model [8] and the UC-GAN model, on overly simplistic and basic substitution ciphers, namely the Caesar cipher, Vigenère cipher, and another basic substitution cipher. To advance beyond this limitation, our research focused on various deep learning architectures, with a special emphasis on the Unified Cipher Generative Adversarial Network (UC-GAN). The initial phase was thus dedicated to assessing UC-GAN's efficacy in decrypting elementary ciphers. This initial evaluation served two main purposes. Firstly, it provided insights into the foundational capabilities of the deep learning UC-GAN model in understanding the fundamental aspects of cryptographic operations. Secondly, it established a baseline for measuring the model's progress in handling more sophisticated cryptographic ciphers which we will test and improve the UC-GAN model to the proposed Unified Multi-Cipher Generative Adversarial Network model (UMC-GAN) with transposition ciphers and rotor ciphers. The UMC-GAN model represents an enhanced version of UC-GAN, expanding its applicability beyond substitution ciphers to include transposition ciphers, such as the rail fence cipher, columnar transposition cipher, and other transposition ciphers. Additionally, UMC-GAN can handle stronger rotor ciphers such as the Enigma, Enigma M4, and Typex rotor ciphers.

While conducting our research, we faced an initial major challenge caused by errors in the provided code and the model's reliance on an NVIDIA GPU and CUDA CuDNN. Resolving this obstacle required approximately three weeks of concentrated effort dedicated to addressing the issues.

A distinguisher was created to evaluate whether it can successfully do the multiclass classification on different ciphers within the same type. It is a separate machine learning model using the CatBoost algorithm [15]. This machine learning algorithm uses gradient boosting on decision trees, to determine which one performed better in the given context. The categorical boosting (CatBoost) algorithm derives its name from its proficiency in handling categorical data ("Cat") coupled with its utilization of gradient boosting techniques ("Boost").

Much like educating a child, the approach to training machine learning models in this study adopts a graduated progression. Initially, these "digital minds" were introduced to the basics of cryptography through substitution ciphers - the equivalent of teaching counting to a child. Once proficiency was established at this foundational level, the models were then challenged with increasingly complex ciphers, such as transposition and rotor ciphers, akin to progressing a child's learning to calculus and integrals. This methodology not only simulates a more natural learning curve but also ensures a comprehensive understanding and adaptability of the models. Testing ranged from simple substitution ciphers to transposition ciphers and then to the sophisticated rotor ciphers, thereby evaluating the deep learning model UC-GAN's ability to generate realistic ciphertexts with different levels of cryptographic complexity.

In this research, our main goal was first to evaluate the capability of the distinguisher in classifying multiple ciphers within the same cipher type. The other goal was to assess the performance of advanced deep learning models, particularly focusing on the unified cipher generative adversarial network model (UC-GAN). We aimed to determine how well the UMC-GAN model could generate realistic ciphertexts for substitution, transposition, and rotor ciphers, and misleading the discriminator. This investigation sought to evaluate the UC-GAN's ability not only to decrypt but also to recognize the nuances presented by cryptographic challenges of varying levels of complexity.

This research advances the field of cryptanalysis by developing the improved UC-GAN model, known as the Unified Multi-Cipher Generative Adversarial Network model (UMC-GAN). UMC-GAN is a deep learning model designed to tackle the complexities of more advanced ciphers without pre-existing knowledge, it addresses the limitations of previous GAN-based cryptanalysis methods, which were constrained by their focus on simplistic cipher types. It focuses on encrypting one plaintext into multiple ciphertexts of the same type and decrypting multiple ciphertexts of the same type into a single plaintext utilizing a single unified generator and a discriminator within its deep learning GAN neural network model. The choice of cipher type, whether substitution, transposition, or rotor ciphers, can be tailored to the specific application. Our study not only challenges the limitations of previous GAN-based cryptanalysis approaches but also establishes a new standard for deep learning applications in decrypting complex ciphers with varying levels of complexity.

The distinguisher achieves an accuracy of around 90% which means it can successfully do the multiclass classification within the same cipher type, namely the substitution, transposition, and the rotor cipher type. The UMC-GAN model reached an accuracy of nearly 98%, which is computed by comparing the generated ciphertexts with the expected ciphertexts. This result indicates that the generator can successfully fool the discriminator by generating realistic ciphertexts. Substitution and transposition ciphers represent classical methods for converting plaintext into ciphertext. With rotor ciphers employed during World War II, enhancing these methods by integrating multiple rotating cipher disks for added encryption complexity and security.

In modern cryptography, these classical methods, namely substitution and transposition, continue to be applied. They embody the confusion property, which obscures the link between the encryption key and its corresponding ciphertext. Confusion is the concept of complicating the association between plaintext and ciphertext through substitution methods. Diffusion is designed to obscure the statistical relationship between the plaintext and the ciphertext. This is commonly achieved via transposition. Modern block ciphers frequently utilize a substitution-permutation network (SPN), where confusion is given by substitution boxes (S-boxes) and diffusion by permutation boxes (P-boxes). The principles of confusion and diffusion in cryptography, from classical methods to modern implementations, highlight their perpetual significance in crafting secure cryptographic systems.

Thus, AI-based cryptanalysis needs to address the properties of confusion and diffusion, especially when attempting to break S-boxes through deep-learning techniques such as GANs. Indeed, the application of GANs in cryptanalysis research is diverse, including Gomez et al.'s work on cracking shift and Vigenere ciphers using the CycleGAN model [8], Ding et al.'s research of private key generation for stream ciphers [5] and Panwar et al.'s survey of end-to-end image encryption using GANs [13]. Furthermore, Gohr et al.'s research on distinguishers for the Speck32/64 lightweight block ciphers using the ResNet model [7] and Baksi et al.'s work on neural distinguishers for lightweight block ciphers such as Gimli, Ascon, Knot, and Chaskey, employing MLP, CNN, and LSTM models [3] are the development of ciphertext distinguishers in modern cryptography.

The core contributions of the research are as outlined. Firstly, we build a CatBoost-based machine learning distinguisher. This distinguisher is designed to evaluate its capability of doing the multiclass classification on different ciphers within the same type. Secondly, we propose the UMC-GAN model, an improved model of UC-GAN. We assess if the deep learning UMC-GAN model can generate realistic ciphertexts. Our research illustrates the UMC-GAN model's capability to execute translations from plaintext to multiple ciphertexts and vice versa. This is accomplished across various ciphers within the same cipher type, including substitution ciphers (Caesar, Vigenère, and generic substitution ciphers), transposition ciphers (rail fence, columnar, and generic transposition ciphers), and rotor ciphers (Enigma, Enigma M4, and Typex rotor cipher), all using a single unified generator and a discriminator of the UMC-GAN model.

Chapter 2

Background

2.1 The GAN Model

The Generative Adversarial Network models (GANs), developed by Ian Goodfellow and colleagues in 2014 [9], is an unsupervised machine learning framework. GANs have shown exceptional abilities in several fields, including converting images from one to another, generating photorealistic images, and reconstructing 3D models from images. Thus, GANs are a state-of-the-art method for creating generative models in the field of deep learning.

There are two neural networks inside the GAN architecture, known as the Generator (G) and the Discriminator (D), which can be viewed as two adversaries in a complex zero-sum game. The generator's work is to create fake samples that closely look like the original samples. The discriminator's role is to identify which samples are real and which are created by the generator, and to estimate how likely each sample is to come from the training data rather than the generator.

The interaction between these two components is regulated by what is known as the standard GAN loss function or the min-max loss, expressed as

$$\min_G \max_D V(D,G) = \mathbb{E}_x[\log D(x)] + \mathbb{E}_z[\log(1 - D(G(z)))] \tag{2.1}$$

where $G(z)$ is the generator's output for a given noise input z, $D(x)$ is the discriminator's probability of x being real data, and $D(G(z))$ is the discriminator's probability, indicates the discriminator's confidence that $G(z)$ is real data, where $G(z)$ is a generated sample produced by the generator. $\mathbb{E}_x$ and $\mathbb{E}_z$ are the average likelihoods of all real data and all generated data respectively.

The standard GAN loss function indicates a competition wherein the generator aims to minimize the value function $V(D,G)$, while the discriminator seeks to maximize it. In the min-max loss equation, the discriminator evaluates a dataset as input, either real (x) or generated ($G(z)$) samples.

In this adversarial mechanism, both the discriminator loss and the generator loss are interconnected, with improvements in one prompting adjustments in the other. It is the evolving relationship and dynamics of these losses that matter, rather than their individual values.

Such adversarial training enables the generator to proficiently create content that appears realistic, successfully deceiving the discriminator. This capability enables the creation of new samples that have a statistical resemblance to the original data.

2.2 Substitution and Transposition Ciphers

Substitution ciphers [1] are cryptographic techniques that replace each letter or symbol in the plaintext with another letter or symbol according to a specific rule. The Caesar cipher, attributed to Julius Caesar, is a monoalphabetic substitution cipher that involves shifting each letter in the plaintext by a fixed number of positions in the alphabet, known as the key. The Vigenère cipher is a polyalphabetic substitution cipher that uses a keyword to create a more complex substitution pattern, making it more secure than the Caesar cipher.

Transposition ciphers [1] rearrange the positions of characters in the plaintext while maintaining the original characters. The rail fence cipher creates a zigzag pattern in which characters are written diagonally and then read off linearly to produce the ciphertext. The columnar transposition cipher rearranges the plaintext by writing it into a grid following a row-wise approach, and then extracting the ciphertext by reading the grid column-wise. The order in which columns are read is determined by the encryption key. These ciphers obscure the message through a rearrangement of characters to enhance the encryption's security level.

2.3 Rotor Machines and Rotor Ciphers

Rotor machines [6] are electromechanical stream cipher devices that have been pivotal in the encryption and decryption of messages throughout much of the twentieth century, contributing significantly to the domain of secure communication. These machines feature a set of rotors, also called wheels or drums, which are rotating disks and have electrical contacts on both sides. The interconnections between these contacts establish a fixed yet complex pattern of monoalphabetic substitution, where each letter is mapped one-to-one with another. Rotor ciphers rely on the rotation of the rotors to perform substitution operations on the plaintext characters, increasing the security of the encrypted messages.

The Enigma machine [16], developed in the early twentieth century, is one of the famous rotor machines. It was extensively used by the German military during World War II for protecting sensitive communications. The operational principle of the Enigma lies in its electromechanical rotor mechanism, which rearranges the 26 letters of the alphabet. In our research, the Enigma machine is equipped with a three-rotor configuration. The Enigma M4 machine was an improved version of the Enigma machine, featuring four rotors and a more complex wiring arrangement, thereby amplifying the difficulty of decryption. On the other hand, the Typex machine [11] was the British response to the Enigma. It was used by the British military to decrypt intercepted German messages. Similar to the Enigma, the Typex machine also employed rotors for encryption, and all variations of Typex machines contained five rotors, compared to the three or four rotors in the Enigma machines.

Chapter 3

Design

3.1 The Distinguisher Model

The objective of the distinguisher is rooted in its definition [17], akin to the concept employed in modern symmetric key ciphers such as the Advanced Encryption Standard (AES), Data Encryption Standard (DES), Blowfish, and International Data Encryption Algorithm (IDEA). In cryptography, a distinguisher is an algorithm designed to discern between the output from a cipher and that of a random process. The designed distinguisher model applied the CatBoost algorithm, a supervised machine-learning algorithm, that aims to determine whether the data under analysis is random (CT_0) or from one of the three specific cryptographic algorithms (CT_1, CT_2, CT_3).

The distinguisher in our research, displayed in Figure 3.1, is tasked with distinguishing among four distinct classes. It has to do the domain classification among four classes (randomly generated texts representing fake ciphertext CT_0, real ciphertext from the first cipher CT_1, second cipher CT_2, and third cipher CT_3).

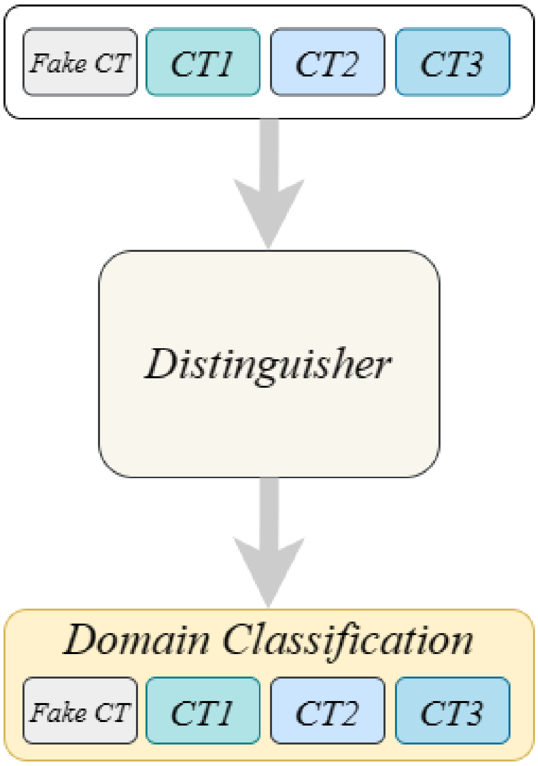


Figure 3.1: The distinguisher model

3.2 The UMC-GAN Model

The UMC-GAN model is an advanced unsupervised deep learning network designed for multi-domain cryptanalysis, building upon the UC-GAN model [14] originally developed for substitution ciphers. The UMC-GAN model is an improved version of UC-GAN because it extends its capabilities to handle various cipher types, including transposition and rotor ciphers, in addition to substitution ciphers.

The objective of both the UC-GAN and UMC-GAN models is to train the generator to work with both plaintext and ciphertext data without pre-existing knowledge. The approach involves utilizing a unified generator, a concept inspired by StarGAN [4], which uses a single unified generator neural network, as shown in Figure 3.2, and a single discriminator neural network, displayed in Figure 3.3, across more than two ciphers of the same type. This training method for handling multiple plaintexts and ciphertexts without prior information is an idea adapted from CipherGAN [8] which requires the use of multiple generators when dealing with more than two distinct ciphers.

Drawing inspiration from StarGAN, the UMC-GAN model applies two neural networks: a discriminator (D) and a generator (G). First, the discriminator learns to distinguish between real and fake texts, for both plaintext and ciphertexts, it also classifies real texts into their respective domains (plaintext PT, ciphertext from the first cipher CT_1, ciphertext from the second cipher CT_2, or ciphertext from the third cipher CT_3) of the same cipher type. Next, the generator takes in both the text and the target domain label, using these inputs to generate a fake text. It attempts to reconstruct the original text from the fake text with the original domain label. The next objective for the generator is to produce texts that are indistinguishable from real texts and can be classified into the target domain by the discriminator. Thus, the UMC-GAN model can perform translations from single plaintext to multiple ciphertexts and vice versa.

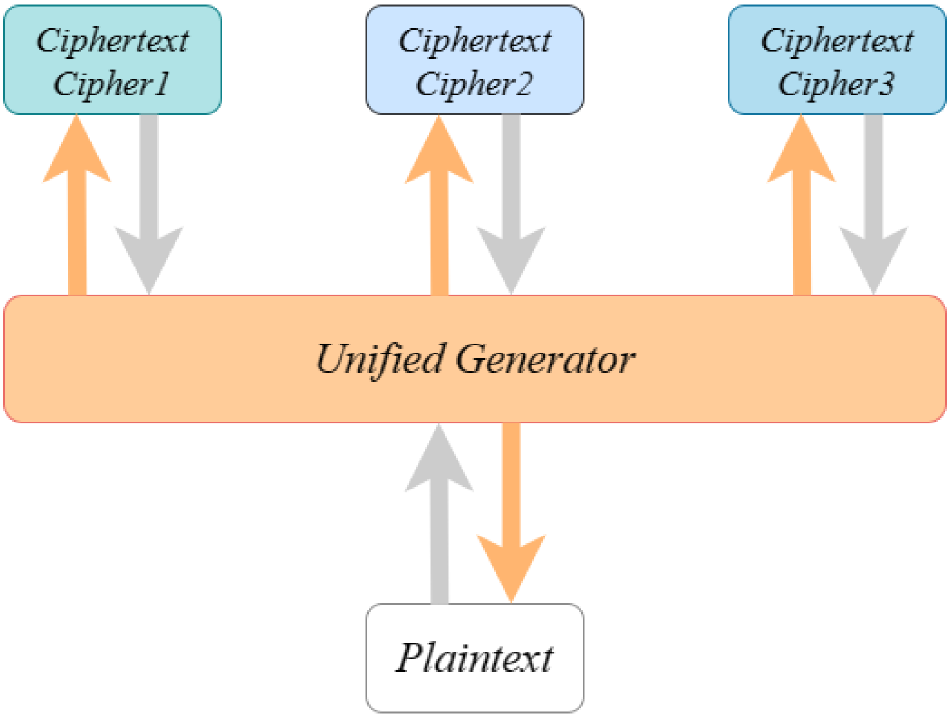


Figure 3.2: UMC-GAN unified generator plaintext generation

The design of the UMC-GAN model's discriminator, referenced in Figure 3.3, is also inspired by the starGAN model, enhancing its capabilities beyond the classical GAN model: to distinguish between real and fake input texts. The advanced discriminator in our research not only differentiates real texts from fake ones but also classifies real texts into their corresponding domains within the same cipher type by introducing an auxiliary classifier [12]. The inclusion of this auxiliary classifier enables the discriminator to manage multiple domains, thereby the UMC-GAN's discriminator can do multiclass classification. During the discriminator's training phase, its first task is to determine the authenticity of the input text, identifying whether it is real or fake. Following this, if the received text is real, the discriminator proceeds to classify its origin, whether it be a plaintext PT, a ciphertext from the first cipher CT_1, the second cipher CT_2, or the third cipher CT_3.

In Figure 3.3, the two arrows directed toward the discriminator are only illustrative. They represent two scenarios: the first case where the input text is real, and the second case where the input text is fake. The discriminator only takes one input text at a time, which can be either real or fake.

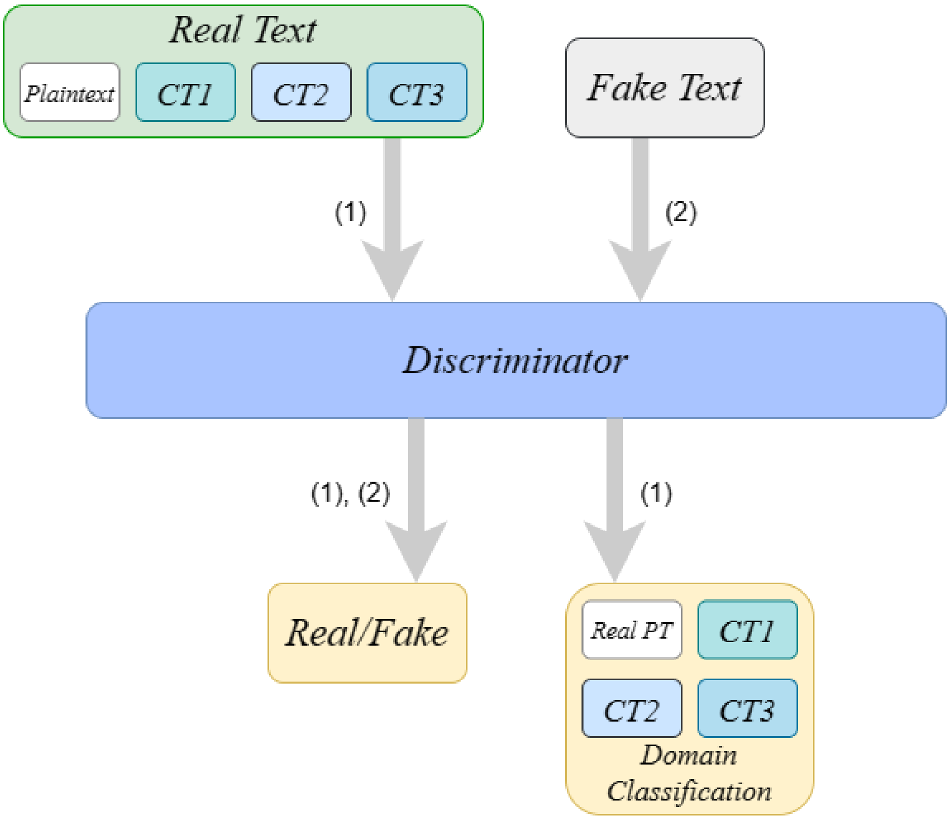


Figure 3.3: UMC-GAN discriminator training[1]

(1) When training with real texts (2) When training with fake texts

[1] PT stands for plaintext. CT stands for ciphertext.

Chapter 4

Implementation

4.1 UMC-GAN and Distinguisher's System Equipment

Table 4.1 provides the system setup details employed for the distinguisher's training.

Table 4.1: System equipment used in distinguisher training.

Component	Description
CPU	Intel Cori7-7700
Language	Python
System type	64-bit operating system
OS type	Window 10/64

The training of the UMC-GAN model was conducted utilizing dual NVIDIA GPU T4 units, as detailed in Table 4.2, a deviation from the initially advised use of dual GTX 1080Ti GPU units in [14]'s UC-GAN model. This adjustment was necessitated due to the unavailability of NVIDIA GPUs in the local setup. Consequently, our training process for both the UC-GAN and UMC-GAN models was executed on Kaggle Notebooks, which afforded us the use of NVIDIA GPU units. This setup also enabled access to CUDA CuDNN.

Table 4.2: System equipment used in UMC-GAN model training.

Component	Description
CPU	Intel Cori7-7700
GPU	NVIDIA GPU T4 x2
Language	Python
System type	64-bit operating system
OS type	Window 10/64

4.2 Neural Architecture of UMC-GAN

The neural architecture of the UMC-GAN model is inspired by that of the UC-GAN model. This neural network is organized into four convolutional blocks. Each block consists of a 1-dimensional (1D) Convolution-Layer Norm-ReLU sequence for channel expansion. Additionally, there is a bottleneck layer that includes five residual layers, and two convolutional blocks that follow the 1D Convolution-Layer Norm-ReLU sequence, are employed to adjust the channel size to 26. Within the discriminator component, a set of three convolutional layers is tasked with extracting features from the recovered plaintexts. Within the generator component, the final layer produces a 26-component vector, and only a single element is highlighted using the Softmax activation function. This highlighted element indicates the alphabet's position. In order to train text domains, each layer in both the generator and discriminator of the UMC-GAN model uses 1-dimensional convolutional layers. The two components of UMC-GAN are further enhanced with layer normalization.

Chapter 5

Evaluation

5.1 Datasets

5.1.1 Dataset for the Distinguisher Model

The dataset in the distinguisher, which uses the CatBoost supervised machine learning algorithm, has 40,000 samples. These samples include 10,000 sets each containing three encrypted plaintexts produced by three distinct ciphers, alongside a random string that represents a fake ciphertext. These data features are classified accordingly: ciphertexts originating from the first, second, and third ciphers are labeled as class 1 (CT_1), class 2 (CT_2), and class 3 (CT_3), respectively. Meanwhile, the randomly generated ciphertext is assigned class 0 (CT_0) as its data label.

The large number of samples is firstly to enhance the model's capacity for generalization, thereby ensuring its efficacy not only on the training data but also on new and unseen datasets, namely the testing and validation datasets. Furthermore, it mitigates the risk of overfitting by providing a wide range of examples. This diversity helps develop a distinguisher model that recognizes general patterns without fitting too closely to the specifics of the training set. Additionally, it can capture subtle and complex patterns, which are often absent in smaller datasets. With only two provided data features, the plaintext and its corresponding ciphertexts: three encrypted plaintexts from three different ciphers and one random string as the fake ciphertext, the large dataset becomes crucial. Moreover, it reinforces statistical techniques employed during the training and validation phases, rendering the model's performance evaluations more reliable.

Once the dataset is created, we split it into three different portions: 80% for training, 10% for testing, and 10% for validation, corresponding to 32,000, 4,000, and 4,000 samples respectively. The training dataset is utilized to develop the model, allowing it to learn and adapt to the data's characteristics. The test dataset, separate from the training set, is used to assess the model's performance, offering insights into how well it generalizes to new, unseen data. The validation dataset, distinct from the training and the test datasets, serves a similar purpose to the test dataset but is often used to perform the final assessment of the model after all adjustments and tuning have been completed. This comprehensive approach ensures a rigorous evaluation of the model's performance.

5.1.2 Dataset for the UMC-GAN Model

The dataset in our research is the Brown Corpus, formally known as the Brown University Standard Corpus of Present-Day American English, which is a digital collection of text samples representing American English.

Our evaluation framework, following the approach employed in the UC-GAN model for evaluation purposes, involves a classification with four distinct labels in both the training and testing datasets of the UMC-GAN and the distinguisher. These labels include the plaintexts (PT) and ciphertexts generated by three different ciphers of the same type (CT_1, CT_2, and CT_3).

Specifically, for substitution ciphers, the labels correspond to the plaintext, the Caesar cipher, the Vigenère cipher, and a generic substitution cipher, aligning with the categorization in the existing UC-GAN model. In the case of transposition ciphers, the labels represent the plaintext, the rail fence cipher, the columnar transposition cipher, and a generic transposition cipher. In the context of rotor ciphers, the labels are designated for the plaintext, the Enigma cipher, the Enigma M4 cipher, and the Typex cipher.

To prepare the dataset for analysis, we perform a standardized pre-processing procedure: we first remove all non-alphabetical characters, including special characters, numbers, punctuation marks, and all whitespaces. After that, the remaining alphabetic characters are converted to lowercase. Finally, to maintain consistency, only the first 100 characters of each text sample are retained.

Hence, whether a sample is a plaintext or a ciphertext, it consistently consists of precisely 100 characters. Within the training dataset for each cipher type - substitution, transposition, or rotor machine - the structure includes 24 samples of plaintext and 72 samples of ciphertext, with the latter including 24 samples for each of the three distinct ciphers of the same type.

5.2 Hyperparameters

5.2.1 Hyperparameters for the Distinguisher Model

The CatBoost algorithm is applied within the distinguisher for a multiclass classification task that involves four distinct classes. This classification scheme is based on the assumption of three ciphers of the same type. In this multiclass classification approach, class 1 is assigned if the distinguisher identifies the ciphertext as originating from the first cipher. If it determines that the ciphertext is encrypted from the second cipher, then the ciphertext is classified as class 2, and similarly, for the third cipher, it is classified as class 3. In cases where the ciphertext is recognized as not belonging to any of the three ciphers after decryption, it is categorized under class 0, indicative of a fake ciphertext.

The hyperparameters employed by the distinguisher, along with their default values for each cipher type, are presented in Table 5.1.

Table 5.1: Hyperparameters and default values for the distinguisher training for various cipher types

Hyperparameter	**Default value for substitution ciphers**	**Default value for transposition ciphers**	**Default value for rotor ciphers**
Number of total iterations for training the distinguisher	500	500	500
Learning rate	0.1	0.1	0.1

5.2.2 Hyperparameters for the UMC-GAN Model

In the evaluation of both the UC-GAN and UMC-GAN models, we adhere to the default hyperparameters established for the UC-GAN model [14]. In this process, we apply the Adam optimizer [10], which is a fast and effective algorithm to optimize stochastic objectives using adaptive estimates of lower-order moments. The hyperparameters specific to the UMC-GAN model, along with those used for its training, are comprehensively shown in Table 5.2 and Table 5.3, respectively, where each parameter's default value is individually listed.

In Table 5.2, the labels' dimension of each cipher type is 4, because there are the plaintext (PT) and the three ciphers' ciphertexts within the same type (CT_1, CT_2, CT_3). In Table 5.3, each epoch equals 1000 iterations.

Table 5.2: Model setting's hyperparameters of UMC-GAN and their default values

Hyperparameter	**Default value**
Dimension of cipher type labels	4
Solver type	rotor Enigma Typex
Number of convolution filters in the first layer of the generator	32
Number of convolution filters in the first layer of the discriminator	32
Weight for domain classification loss λ_{cls}	1
Weight for reconstruction loss λ_{rec}	10
Weight for gradient penalty	10

Table 5.3: Training setting's hyperparameters for UMC-GAN and their default values

Hyperparameter	**Default value**
Batch size	32
Number of total iterations for training the discriminator and the generator	1,060,000
Learning rate for the generator	0.00018
Learning rate for the discriminator	0.00018
Beta1 for Adam optimizer	0
Beta2 for Adam optimizer	0.9

5.3 Distinguisher Performance on Various Cipher Types with Hyperparameter Optimization

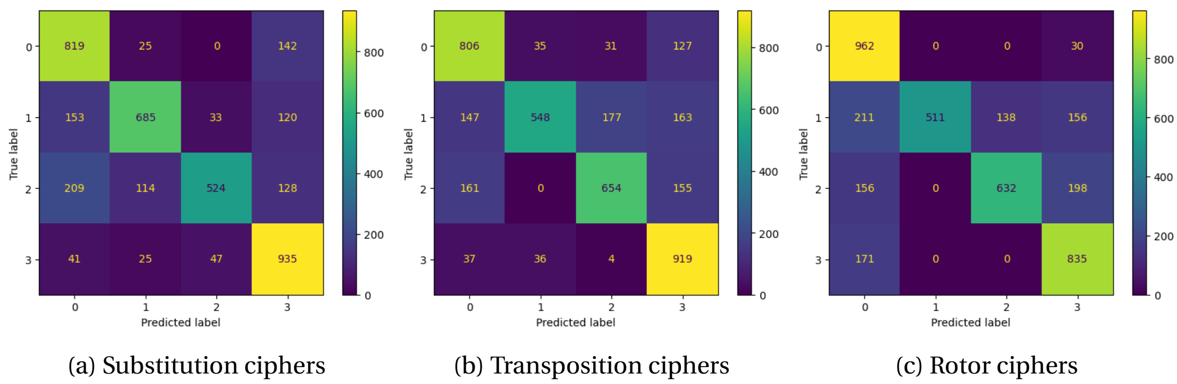
The developed distinguisher, utilizing the CatBoost algorithm, aims to distinguish among four classes: ciphertexts generated by three different ciphers (CT_1, CT_2, CT_3), and a fake ciphertext (CT_0), as described in Section 3.1. During the hyperparameter optimization process, various hyperparameter configurations were tested on the distinguisher to determine the optimal setting. For this process, a dataset comprising N = 40,000 samples was employed for training, testing, and validation purposes. The experimentation involved learning rates of 0.1, 0.05, and 0.01, along with a total iteration number of 500 and 700. Table 5.1 outlines the default hyperparameter setting used for training the distinguisher, and Table 5.4 displays the optimal hyperparameter configuration identified for training the distinguisher with N = 40,000 samples.

The learning rate set initially in the CatBoost model configuration can differ from the actual learning rate that is used during the training. The actual learning rate is influenced by factors including the number of iterations and the characteristics of the input dataset. CatBoost adopts a dynamic learning rate strategy, automatically adjusting the learning rate according to the training's progression and the mechanism for detecting overfitting. This dynamic adjustment enhances the model's accuracy and reduces its training time. Therefore, the actual learning rate detailed in Table 5.4 represents the optimal learning rate determined through this dynamic process, which is applied in the model training.

Table 5.4: Optimal hyperparameter tuning for the distinguisher training process when N = 40,000

Hyperparameter	**Substitution ciphers**	**Transposition ciphers**	**Rotor ciphers**
Number of total iterations for training the distinguisher	700	500	500
Given learning rate	0.05	0.1	0.1
Actual learning rate used	0.132618	0.14722	0.14722

In order to evaluate the performance of the machine-learning-based distinguisher, confusion matrices referenced in Figure 5.1 were constructed with N = 40,000 samples for each cipher type. Classification metrics including true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) are displayed in the confusion matrices. Additionally, performance indicators such as accuracy, precision, recall, F1 score, macro F1 score, and weighted F1 score, detailed in Tables 5.5, 5.6 and 5.7, serve as evaluation measures for assessing the performance of classification models.



(a) Substitution ciphers (b) Transposition ciphers (c) Rotor ciphers

Figure 5.1: Confusion matrix for each cipher type when N = 40,000

Table 5.5: Classification metrics for the distinguisher with substitution ciphers

Metric	Fake ciphertext	Caesar	Vigenère	Substitution
Accuracy	0.8575	0.8825	0.8672	0.8742
Precision	0.6702	0.8068	0.8675	0.7057
Recall	0.8306	0.6912	0.5374	0.8922
F1 score	0.7418	0.7446	0.6637	0.7880
Macro F1 score	0.7345			
Weighted F1 score	0.7356			

Table 5.6: Classification metrics for the distinguisher with transposition ciphers

Metric	Fake ciphertext	Rail fence	Columnar	Transposition
Accuracy	0.8655	0.7003	0.8068	0.7498
Precision	0.8605	0.8853	0.5295	0.6626
Recall	0.8680	0.7552	0.6742	0.7124
F1 score	0.8695	0.6738	0.9227	0.7788
Macro F1 score	0.7259			
Weighted F1 score	0.7254			

Table 5.7: Classification metrics for the distinguisher with rotor ciphers

Metric	Fake ciphertext	Enigma	Enigma M4	Typex
Accuracy	0.8580	0.8738	0.8770	0.8612
Precision	0.6413	1.0000	0.8208	0.6850
Recall	0.9698	0.5030	0.6410	0.8300
F1 score	0.7721	0.6693	0.7198	0.7506
Macro F1 score	0.7279			
Weighted F1 score	0.7277			

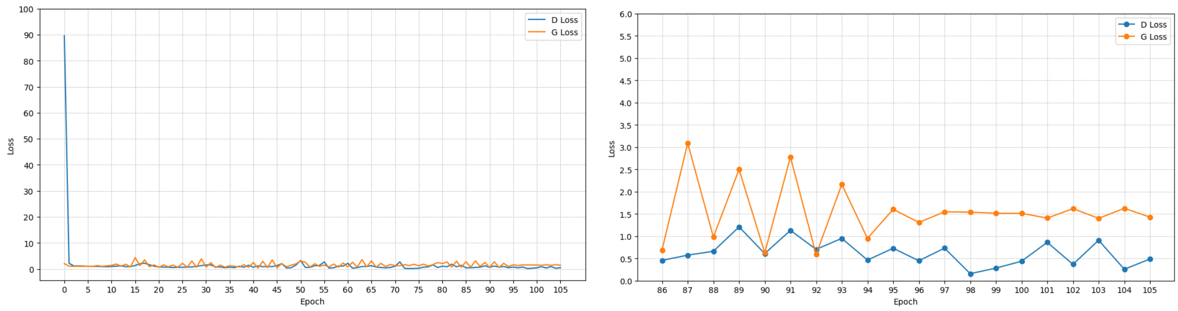
We evaluated the accuracy of the ability that the distinguisher is able to distinguish between the classes CT_0, CT_1, CT_2, and CT_3. The data in Tables 5.5, 5.6 and 5.7 show that with the optimal hyperparameter setting, the accuracy is around 90%. Therefore, the distinguisher successfully classifies the ciphertexts into four classes, namely the fake ciphertext and the ciphertexts from the first, second, and third ciphers.

5.4 UMC-GAN Performance on Various Cipher Types

In the Generative Adversarial Networks (GANs) adversarial setup, there exists a reciprocal relationship between the losses of the discriminator and the generator: enhancements in one often lead to changes in the other. The discriminator and generator losses, the loss metrics in GANs, depend on the dynamics of the adversarial interplay between these two components. Ideally, the discriminator loss should be close to zero, indicating that it can correctly distinguish and classify input data. Conversely, a high discriminator loss suggests either deception by the generator or that the discriminator is too confident and rejects real data. The generator loss is expected to be high, signifying its ability to effectively deceive the discriminator with fake data it generates. A low generator loss implies a failure to produce convincing fake data. For a GAN to be considered stable, these loss metrics of the GAN model should converge to stable values, signifying an equilibrium where the model cannot make further improvements.

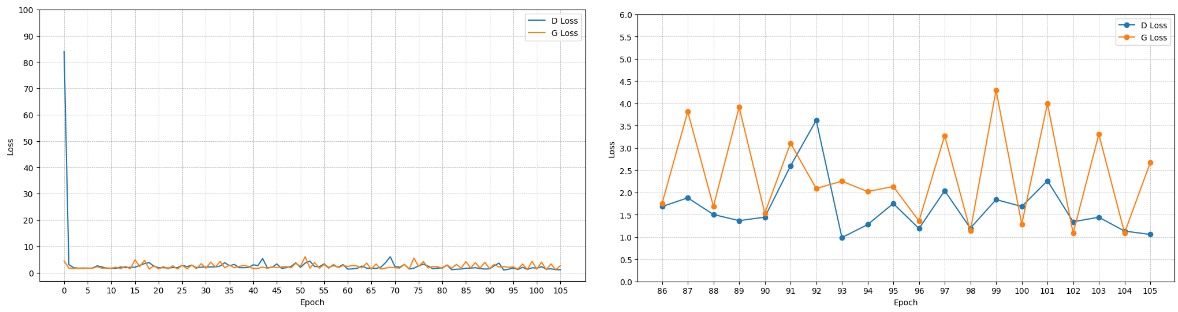
In the experimental result of UMC-GAN, the discriminator loss and generator loss graphs for each cipher type, as depicted in Figures 5.2a, 5.3a, and 5.4a , demonstrate a convergence toward stable values, which indicates that UMC-GAN is stable. Additionally, the performance of the UMC-GAN model was verified by examining the generated samples and computing the model's accuracy. It is computed by comparing the generated with the expected (target) ciphertexts.

Figures 5.2b and 5.4b, which are the detailed versions over the last 20 epochs of Figures 5.2a and 5.4a for substitution and rotor cipher types, respectively, reveal that the discriminator loss is close to zero. This means the discriminator of the UMC-GAN model correctly classifies the input texts. The generator exhibits a relatively higher loss, as seen in the graphs of Figures 5.2b and 5.4b. These metrics suggest proficient learning by the UMC-GAN model in these cipher types. This means that the generator effectively fools the discriminator. Indeed, the accuracy results are nearly 98% as shown in Figures 5.5a and 5.5c over 106 epochs.



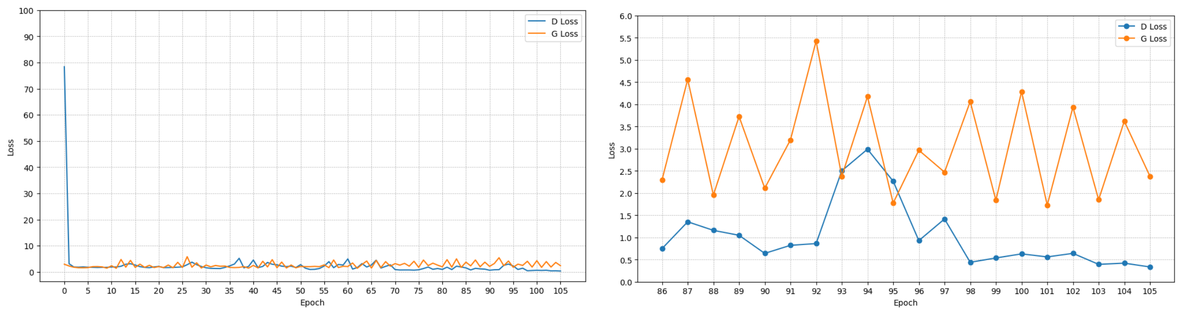
(a) (b)

Figure 5.2: Discriminator loss (D Loss) and generator loss (G Loss) over
(a) 106 epochs (b) the last 20 epochs
for substitution ciphers



(a) (b)

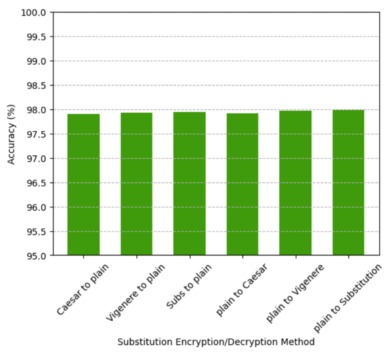
Figure 5.3: Discriminator loss (D Loss) and generator loss (G Loss) over
(a) 106 epochs (b) the last 20 epochs
for transposition ciphers



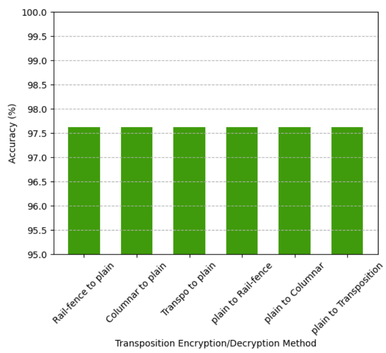
(a) (b)

Figure 5.4: Discriminator loss (D Loss) and generator loss (G Loss) over
(a) 106 epochs (b) the last 20 epochs
for rotor ciphers

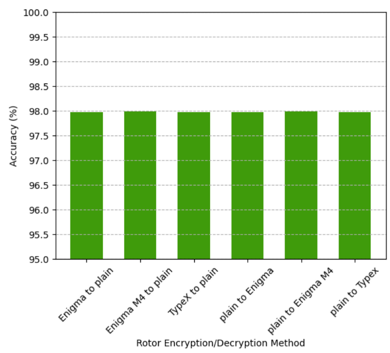
In Figure 5.3b, the precise view of Figure 5.3a, which focuses on the last 20 epochs of the UMC-GAN training in the transposition cipher shows the discriminator loss value consistently exceeding one. The generator loss values occasionally match or fall below the discriminator loss values in the last 10 epochs. This pattern indicates the discriminator can be easily fooled by the generator and the generator is not able to generate realistic data. Despite its instability, the UMC-GAN model maintains, approximately 97.5%, almost 98% accuracy in this cipher type as well, as demonstrated in Figure 5.5b over 106 epochs.



(a) Substitution ciphers



(b) Transposition ciphers



(c) Rotor ciphers

Figure 5.5: Accuracy for each cipher type over 106 epochs

The UMC-GAN's performance is evaluated based on the accuracy computed by comparing the generated with the expected ciphertexts. The experimental results indicate that the UMC-GAN model is capable of generating realistic and diverse ciphertexts for substitution, transposition, and rotor ciphers.

Therefore, in the context of plain-to-multi-ciphers translation, the UMC-GAN model is able to reconstruct ciphertexts of the same type from a single plaintext and vice versa in multi-cipher-to-plain translation. UMC-GAN can successfully decrypt and break all the ciphers of the same type using a single unified generator.

Chapter 6

Discussion

We built a distinguisher based on CatBoost was developed to evaluate whether it can do the multiclass classification on different ciphers within the same type well. We also introduced the Unified Multi-Cipher Generative Adversarial Network model (UMC-GAN), an improved version of the Unified Cipher Generative Adversarial Network model (UC-GAN). This model is capable of performing translations from multi-ciphertexts to a single plaintext and vice versa across various cipher types, utilizing one single unified generator and a discriminator within its deep learning GAN neural network model. The UMC-GAN can effectively break substitution, transposition, and rotor ciphers, without pre-existing knowledge.

In our experimental results, the accuracy of the distinguisher consistently remained near 90% for all cipher types. Therefore, the distinguisher can successfully do the multiclass classification on different ciphers within the same cipher type - substitution, transposition, or rotor cipher type. The UMC-GAN model reached close to 98% accuracy, computed by comparing the produced ciphertexts with the expected ones. This result indicates that the generator can successfully fool the discriminator by generating fake but realistic texts.

Our work pioneers a new category of AI-based attacks on substitution, transposition, and rotor ciphers, employing unsupervised deep learning techniques. The UMC-GAN model also presents a new approach for generating and transforming ciphertexts across different domains and cipher types using a unified generative neural network.

Beyond the UMC-GAN model's ability to decrypt ciphers back to World War II, it holds potential for cracking modern block ciphers and for applications in other fields that require multi-to-single or single-to-multi-domain translations. This includes generating coherent unified image or voice outputs from multiple image or voice inputs, using the UMC-GAN model.

Chapter 7

Conclusion

In conclusion, we first built a CatBoost-based distinguisher to evaluate if it can successfully do the multiclass classification on different ciphers within the same type. The distinguisher's accuracy is around 90% for all cipher types, so it can classify multiple cipher domains of the same cipher type. We also proposed the Unified Multi-Cipher Generative Adversarial Network model (UMC-GAN), a model capable of converting a single plaintext to multiple ciphertexts and multiple ciphertexts to a single plaintext, not only across substitution ciphers but also across transposition and rotor ciphers. This is achieved using only one unified generator and a discriminator in the deep learning GAN neural network model. The UMC-GAN model has demonstrated its proficiency in decrypting a variety of cipher types, even without prior knowledge of these ciphers. To evaluate its efficacy of generating realistic ciphertexts, we computed its accuracy by comparing the generated ciphertexts and the expected ciphertexts. The UMC-GAN model's performance was remarkable, attaining an accuracy of nearly 98%. This high accuracy implies that the generator is effective in producing realistic ciphertexts for substitution, transposition, and rotor ciphers, and misleading the discriminator. Hence, the UMC-GAN model is able to decrypt ciphers within the same type.

In the future, we aim to explore the UMC-GAN model performance on mixed cipher types. We also aim to generate realistic ciphertexts under lightweight block ciphers including Speck32/64 and Ascon, as well as those employed in practical applications such as Telegram and WhatsApp.

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